Question Paper

Exam Date & Time: 03-Jan-2026 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026
SUBJECT: CSS 2101/CSS_2101 DATA STRUCTURES
(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) A bank wants to manage customer account numbers efficiently using an AVL tree to ensure fast access and updates. The following 4-digit account numbers need to be inserted into the AVL tree in this order: 1420, 1300, 1250, 1310, 1500, 1550, 1530. (5)
i) Insert the account numbers one by one into the AVL tree, performing rotations whenever the tree becomes unbalanced.
ii) After all insertions, delete the account numbers 1530 and 1420 from the tree, rebalancing as necessary.
At each step (after every insertion and deletion), show the balance factor of each node.

Course Outcome: "CO4" / Cognitive Level: "Application"

1B) You are required to implement four stacks in an array of size 20. Assume the following initial declarations. (3)
element memory[20]; // Global memory declaration
int top[4];
int boundary[4];
Answer the following.
i) Write the C code snippet required to initialize the top and boundary of each stack.
ii) Develop the C function to push an element into the ith stack.
iii) Develop the C function to pop an element from the ith stack.

Course Outcome: "CO3" / Cognitive Level: "Application"

1C) Write a C program to read and display employee details including the date of joining using nested structure. Consider date of joining as nested structure. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

2A) Write a function to convert an infix expression to postfix. Include the following functions as per the prototype. (5)
i) int InStackPrecedence(char op) ; to return the precedence of operator within the stack
ii) int IncomingPrecedence(char op); to return the precedence of the incoming operator.
iii) void Infix_Postfix(char infix[], char postfix[]); to convert given infix expression to postfix.
Main function not required.
Note: The infix expression may include ONLY the following operators (listed in the order of their precedence):
• ^ (exponentiation) → **right associative**
• * (multiplication), / (division) → **left associative**

• + (addition), - (subtraction) → **left associative**

Course Outcome: "CO3" / Cognitive Level: "Application"

2B) i. Write a C function to perform a level-order traversal for the binary tree. (3)
ii. Determine the inorder, preorder, and postorder traversal node sequences for the given binary tree in Fig.Q.2B.

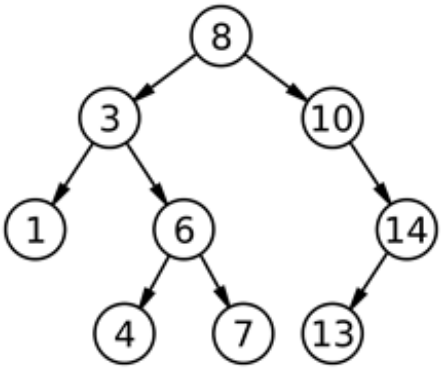


Fig.Q.2B

Course Outcome: "CO4" / Cognitive Level: "Application"

2C) Write a C program to dynamically allocate memory for 2D string arrays (array of strings) and perform concatenation using pointers. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

3A) i) Write a function to perform the Transpose computation of a given sparse matrix using linked list. (5)
ii) Demonstrate complete step-by-step process to compute the transpose of the given in Fig.Q.3A sparse matrix using the above function.

$$\begin{bmatrix} 0 & 2 & 0 & 3 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 & 0 & 5 \\ 6 & 0 & 9 & 0 & 0 & 0 \end{bmatrix}$$

Fig.Q.3A

Course Outcome: "CO1" / Cognitive Level: "Application"

3B) In a gaming leaderboard system, players' scores are maintained in a Binary Search Tree (BST) for efficient ranking and updates. Write an iterative function to delete a player's score from the BST when the player leaves the game or their record needs to be removed. (3)
Delete the score 30 from the BST given below and illustrate the resulting tree structure after deletion.

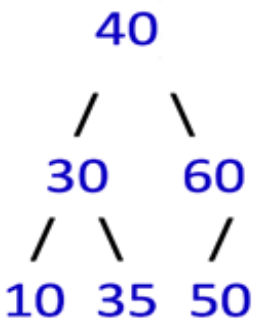


Fig.Q.3B

Course Outcome: "CO4" / Cognitive Level: "Application"

3C) Write a C function bool isDuplicate(char s[]) that detects duplicate parentheses in a given expression using a stack. (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

4A) A mobile app named **"MIT-BLR Connect"** maintains contact details of alumni in three different modules: (5)

i) **Add Recent Contacts Module:** Displays recently interacted alumni in order of contact - new ones added at the beginning.

ii) **Traverse Module:** Allows viewing alumni details in forward direction.

iii) **Develop a module to print the data in reverse order.**

Develop C functions considering singly linked list as the suitable data structure

Course Outcome: "CO2" / Cognitive Level: "Analysis"

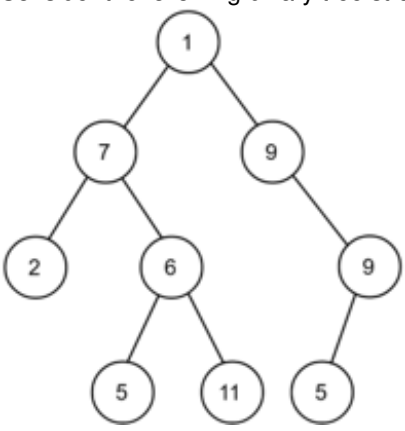
4B) Consider the following expression to construct an expression tree. (3)

$$(P + Q) \times (R - S/T)$$

Provide the sequence of symbols generated by performing a **pre-order** traversal of this specific expression tree.

Course Outcome: "CO4" / Cognitive Level: "Application"

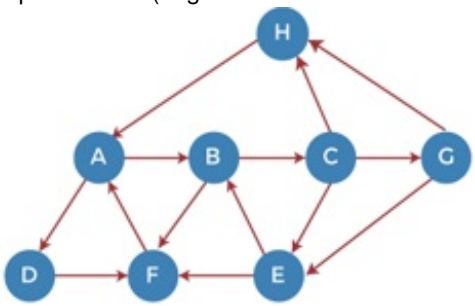
4C) Consider the following binary tree structure (2)



Identify all the **leaf nodes** in the provided tree structure. Develop a C function for counting the number of leaf nodes.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

5A) Consider the directed graph shown below, representing a set of tasks A,B,C,D,E,F,G,H and their dependencies (edges indicate direction of dependency). (5)



Consider the source vertex as **A**. The traversal order should be alphabetical order.

i) Construct **Adjacency List** for the given graph by inserting the adjacent nodes in alphabetical order.

ii) Show the **order of traversal** of vertices using DFS.

iii) Develop recursive C function for performing DFS.

Check the existence of spanning tree.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

5B) A hospital wants to maintain a dynamic list of patients waiting for a doctor. (3)
Since the number of patients keeps changing (patients arrive and get treated at different times), a **Doubly Linked List** is chosen to manage this queue efficiently.
Each node in the list should store:
• Patient ID
• Patient Name
Design the C functions for the following operations:
i) **Adding** a new patient at the end of the list.
ii) **Removing** a patient from the front (when treatment is done).
Displaying the current waiting list.

Course Outcome: "CO2" / Cognitive Level: "Application"

5C) A **Priority Queue** is a special type of queue where each element is associated with a priority. (2)
Elements with higher priority are dequeued before elements with lower priority, even if they were added later.
(a) Explain the concept of a **Priority Queue** and how it differs from a regular queue.
(b) Given the following elements with priorities:
(10, 3), (20, 1), (30, 2) (where the first value is the element, and the second value is the priority).
Show the order in which these elements would be dequeued from the priority queue, assuming **higher priority** means smaller priority number.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSS_2101:Data Structures

#	Course Outcome	Description	Marks
1	1	CO 1	9
2	2	CO 2	8
3	3	CO 3	12
4	4	CO 4	16
5	5	CO 5	5
		Total	**50**

* NA - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSS_2101:Data Structures

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	34(68%)
4	Analysis	5	16(32%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)